RTDL: A Python-Hosted Ray-Tracing DSL for Non-Graphical Spatial Workloads

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Abstract—Ray tracing (RT) is widely used for image synthesis because it solves hierarchical geometric search efficiently. The same capability is attractive for non-graphical workloads such as spatial joins, point-in-polygon tests, and geometric candidate generation, but direct use of RT runtimes still requires backend-specific engineering. We present RTDL (Ray-Tracing Domain Language), a Python-hosted DSL and compiler/runtime system for expressing such workloads once and lowering them to multiple backends while preserving explicit semantics. RTDL separates authored kernel meaning from backend-specific execution details such as acceleration-structure construction, launch state, buffer layout, and result materialization. We evaluate RTDL on the RayJoin workload family using stable real-data packages and controlled scalability studies. Across the accepted package surfaces, RTDL preserves exact row-level parity among a native C++ oracle, Embree, OptiX, and an external PostGIS reference. On the main long County–Zipcode point-in-polygon workload reported in this paper, RTDL with Embree and RTDL with OptiX both outperform PostGIS on the published prepared-execution and repeated-call boundaries. The result is a reproducible, multi-backend evaluation showing that a programmable RT language/runtime stack can be a credible way to build non-graphical ray-tracing workloads.

Index Terms—domain-specific language, ray tracing, spatial joins, Embree, OptiX, PostGIS, RayJoin, reproducibility

I. Introduction

The recent proliferation of hardware-accelerated ray-tracing (RT) cores and high-performance traversal kernels has transformed graphics, but their potential for non-graphical spatial computing remains underexploited. Operations such as spatial joins, point-in-polygon tests, and geometric candidate filtering are natural fits for BVH-style traversal and intersection machinery. In practice, however, exploiting that hardware still requires backend-specific engineering around memory layout, launch configuration, payload conventions, and numerical boundary semantics. The programming burden is especially high when the goal is to preserve the same workload semantics across CPU and GPU execution paths.

RTDL is intended to reduce that programming burden by raising the authoring surface from backend-specific APIs to a compact kernel DSL while retaining multiple execution backends. The language is hosted in Python, but its purpose is not scripting convenience alone. Its purpose is to let an author specify geometric inputs, traversal intent, refine predicates, and output rows once, then lower that kernel into controlled runtime paths with comparable semantics. The central question of this paper is whether such an abstraction can remain useful and defensible when evaluated on a demanding, already-published workload family.

RayJoin is the right target for that evaluation because it exercises exactly the class of workloads RTDL is meant to support: non-graphical ray-tracing queries over spatial data with both correctness and performance demands. This paper therefore does two things. First, it presents RTDL as a language/runtime design for this workload class. Second, it evaluates RTDL on a carefully scoped RayJoin-oriented package consisting of stable real-data cases and controlled scaling studies. The scope is intentionally narrower than a full paper-identical reproduction: unavailable dataset families remain deferred, and the current overlay result remains a seed-generation surface rather than full polygon materialization.

This paper makes the following contributions:

1) The RTDL System: We introduce a Python-hosted DSL and compiler that lowers high-level spatial kernels into a backend-independent intermediate representation, separating kernel semantics from backend-specific traversal and scheduling.
2) Multi-Backend Runtime: We describe an extensible runtime architecture that supports execution across Intel Embree (CPU), NVIDIA OptiX (GPU), and a native C++ semantic oracle.
3) Cross-System Correctness Methodology: We establish a rigorous validation framework that ensures exact, row-level parity across heterogeneous execution paths, utilizing PostGIS as an external industrial baseline for real-data packages.
4) RayJoin-Oriented Evaluation: We provide a reproducible, multi-backend evaluation of RayJoin-style LSI, PIP, and overlay-seed workloads on stable datasets, demonstrating that RTDL can preserve exact semantics while targeting heterogeneous RT backends.

II. Related Work

A. Spatial Join Systems

RayJoin is the closest direct comparison point for this paper because RTDL is explicitly evaluated against its workload family, artifact structure, and reporting surfaces [1]. PostGIS remains the most important external reference in the current RTDL evaluation because it provides a widely used industrial spatial database with

indexed spatial predicates and well-understood geometry semantics [2]. GeoSpark SQL represents a different design point: it exposes SQL-level spatial operations on top of a distributed Spark/Hive execution substrate rather than a backend-portable kernel language [3]. Other widely used distributed spatial systems, such as SpatialHadoop and Apache Sedona, occupy the same broad system class even though they expose different storage and execution tradeoffs [4], [5]. RTDL differs from all of these classes of systems. It is not a database, and it is not a paper-specific one-off implementation. Instead, it aims to express non-graphical ray-tracing workloads through a constrained kernel language and then lower them into multiple execution backends while keeping explicit correctness contracts.

B. Compiler and DSL Context

RTDL also relates to compiler-backed domain-specific systems such as Halide [6] and TVM [7]. Those systems separate high-level computation intent from lower-level schedule or backend realization and are therefore important precedents for RTDL's own authored-kernel to backend-plan split. The difference is domain focus. Halide and TVM center image-processing and tensor-operator pipelines, whereas RTDL centers geometric candidate generation, refine predicates, and emitted join rows. The relevant design pressure is therefore not only optimization, but exact spatial semantics across heterogeneous backends.

C. Ray-Tracing Runtime Context

At the runtime layer, RTDL builds on ray-tracing infrastructure rather than reimplementing traversal from scratch. Embree provides the validated CPU ray-tracing kernel framework [8], while OptiX provides the reported GPU execution path in this paper [9]. RTDL's contribution is not a new traversal engine. The contribution is the language/runtime boundary, the native oracle, and the reproducible multi-system evaluation built on top of these runtimes.

III. What Ray Tracing Is and Why It Matters Beyond Graphics

In graphics, ray tracing answers visibility questions by traversing geometric hierarchies quickly. Rays, bounding-volume hierarchies, and fast intersection kernels make it practical to search complex geometric scenes without checking every primitive pair directly. The same basic pattern also appears in non-graphical workloads. A spatial join, for example, can often be expressed as a broad-phase search for candidate geometry pairs followed by an exact refine step on the surviving candidates.

This observation is what makes ray tracing interesting for spatial computing. Workloads such as line-segment intersection, point-in-polygon, overlap candidate generation, and geometric counting all involve the same kind of hierarchical search that RT systems already accelerate well. Systems such as RayJoin [1] demonstrate that this mapping is not merely conceptual: ray-tracing cores can be used effectively for spatial join work.

The problem is that using RT backends directly still exposes a great deal of backend-specific engineering. Developers must reason about acceleration structures, payload/state conventions, memory layout, launch details, and boundary behavior separately for CPU and GPU paths. RTDL addresses that problem instead of trying to replace the underlying traversal engines.

A. RayJoin as the First Application Target

RayJoin is the right first target for RTDL because it exercises the exact class of workloads RTDL is meant to support [1].

The current RTDL paper does not claim a paper-identical reproduction of every RayJoin dataset family. Instead, it uses RayJoin as the workload and evaluation reference point for a more language/runtime-oriented question: can one authored kernel surface survive lowering to multiple RT backends while preserving exact semantics and producing honest performance claims?

The evaluation surface mirrored from RayJoin includes three workload families:

- LSI: line-segment intersection style joins,
- PIP: point-in-polygon style joins,
- Overlay: currently represented in RTDL as an overlay-seed evaluation, which identifies intersecting polygon pairs without full geometry materialization.

For this work, RayJoin Figure 13 and Figure 14 are represented through controlled synthetic studies, while RayJoin Table 3, Table 4, and Figure 15 are represented through validated results on stable dataset packages.

IV. Why RTDL and How It Works

A. Why a DSL Is Needed

The central design claim of RTDL is that non-graphical RT workloads should be programmable at the workload level rather than at the backend-API level. Without such an abstraction, every new workload must be translated manually into backend-specific setup code, traversal configuration, payload plumbing, buffer layout, and result marshaling. That makes correctness difficult to compare across systems and slows down experimentation.

RTDL therefore tries to keep the authored surface close to the workload itself:

- which geometries are inputs,
- which side is probe or build,
- what traversal is intended,
- what refine predicate defines final truth,
- which fields belong in the emitted rows.

The backend-specific execution details are still present, but they move below the language boundary instead of staying in every authored workload.

B. Language Surface

RTDL is authored from Python through a constrained kernel DSL. A user writes a kernel by declaring typed inputs, traversal intent, refine predicates, and output fields. The intent is that authors specify workload meaning rather than backend launch details. The same language surface currently supports six implemented workloads in the repository, but the RayJoin-facing slice of this paper focuses on LSI, PIP, and overlay.

C. Representative Kernel

Listing 1 shows the shape of an RTDL PIP kernel. The author declares probe/build inputs, asks for BVH traversal, selects a refine predicate, and specifies the emitted row schema. Backend-specific details such as Embree scene construction, OptiX launch state, and row marshaling remain outside the authored kernel.

Listing 1. Representative RTDL PIP kernel.

```
@rt.kernel(backend="rtdl", precision="float_approx")
def point_in_counties():
    points = rt.input("points", rt.Points, role="probe")
    polygons = rt.input("polygons", rt.Polygons, role="build")
    candidates = rt.traverse(points, polygons, accel="bvh")
    hits = rt.refine(
        candidates,
        predicate=rt.point_in_polygon(exact=False,
                                      boundary_mode="inclusive"),
    )
    return rt.emit(hits, fields=["point_id", "polygon_id", "contains"])
```

D. Compiled Representation and Lowering

RTDL lowers authored kernels into a compiled-kernel representation that separates language semantics from backend realization. The lowering path remains RayJoin-oriented in the current codebase: it maps geometry roles, candidate generation, refine predicates, and emitted rows into backend-specific execution plans. In practice, the lowered form records the probe/build role assignment, the traversal primitive to invoke, the refine predicate family, precision and boundary-mode settings, and the emitted row schema. The backend adapters then translate that representation into concrete execution state such as Embree scenes, OptiX launch parameters, native buffers, and row marshaling logic. This allows the same kernel to execute against a semantic oracle, a CPU Embree runtime, and a GPU OptiX runtime without changing the authored kernel surface.

E. Execution Backends

The evaluated backends for this paper are:

- Native C/C++ oracle: the semantic ground truth for the reported results.
- Embree: the primary validated CPU backend.
- OptiX: the validated GPU backend on the Linux workstation in Table I.

The Vulkan backend remains in the repository, but the paper's main performance story focuses on Embree and OptiX because those are the mature high-performance backends on the accepted long exact-source surface.

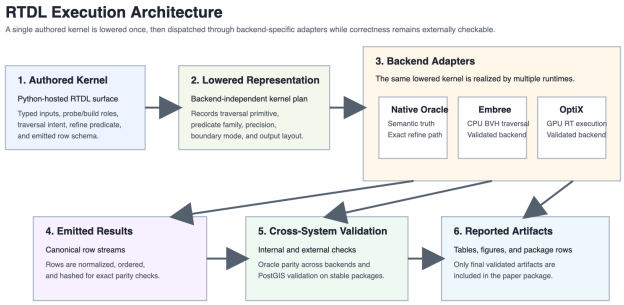

Fig. 1. RTDL execution architecture. A single authored kernel is lowered into a backend-independent representation, dispatched through backend adapters, and checked through emitted-row parity and external validation.

F. Execution Pipeline

The execution path is conceptually simple even though the backend implementations are not. RTDL first compiles the authored kernel into a backend-independent representation that records geometry roles, traversal structure, refine predicate, and emitted fields. The runtime then normalizes input geometry, invokes the selected backend, and converts the returned native rows into the requested user-facing representation. The crucial discipline in this paper is that all reported real-data results are checked against the native oracle and, for the validated real-data packages, against PostGIS as an external reference.

This separation between authored kernel, lowered representation, and runtime adapter is the main systems contribution of RTDL. It keeps semantic decisions visible at the language level while still allowing backend-specific execution strategies underneath. The resulting paper is therefore not just a benchmark of Embree or OptiX; it is an evaluation of whether one authored workload surface can survive translation into multiple RT backends without drifting semantically.

V. Design Considerations and Evaluation Methodology

A. Design Considerations

Three design considerations drive the current RTDL package.

First, semantic consistency is more important than backend-local cleverness. The paper compares systems

that execute related geometric work under different runtime conditions, so the language/runtime layer must make boundary modes, refine predicates, and emitted rows explicit enough to compare honestly.

Second, RTDL must separate broad-phase candidate generation from final truth. That principle shaped the mature positive-hit point-in-polygon paths reported here. Embree and OptiX both use backend-specific candidate generation, but the reported accepted paths keep final inclusive truth on the host through exact finalization.

Third, the Python host is allowed to orchestrate but should not repeatedly do bulk geometry work on the hot path. The strongest current performance wins only appeared after the runtime began reusing prepared executions and prepacked geometry instead of rebuilding them unnecessarily.

TABLE I
Execution environment for the reported four-system package.

Item	Value
Host class	Linux validation workstation
OS	Ubuntu 24.04.4 LTS
CPU	Intel Core i7-7700HQ
Memory	16 GiB installed (about 15 GiB usable)
GPU	NVIDIA GeForce GTX 1070
Driver	580.126.09
CPU RT backend	Embree 4.3.0
GPU RT backend	OptiX 9.0.0
External checker	local PostgreSQL/PostGIS on the same host
GEOS role	exact prepared-polygon covers(point) path for validated PIP parity

B. Optimization Strategy

The optimization story in this paper is therefore not "replace PostGIS with a single GPU kernel." It is a layered systems story.

At the backend level, RTDL relies on mature traversal runtimes such as Embree and OptiX. At the language/runtime level, the main optimizations are:

- over-approximating backend candidate generation with host exact finalize,
- prepared-execution reuse,
- canonical-record and cached-prepared fast paths,
- reuse of prepacked point and polygon buffers,
- moving one-time launch costs out of repeated timed prepared execution surfaces when that boundary already charges them to preparation.

These optimizations matter because the practical performance gap in a DSL system often comes not from raw traversal speed, but from repeated packing, binding, and result-materialization overhead around the backend itself.

C. Correctness Method

The correctness methodology is intentionally layered. The native C/C++ oracle serves as the internal semantic ground truth. Embree and OptiX are checked against that oracle. For validated real-data packages, PostGIS serves as an external industrial reference. PostGIS queries are executed with indexed GiST-assisted plans rather than brute-force SQL. For PIP, PostGIS positive-hit queries are expanded into the same full-matrix RTDL truth semantics before parity comparison.

The PIP parity path used in this paper is also more specific than a generic manual point-in-polygon implementation. During development, a systematic external comparison against PostGIS exposed remaining boundary/-topology mismatches, so the RTDL refine path adopted GEOS-backed prepared-polygon covers(point) semantics for the native oracle and Embree, and the reported OptiX overlay result was observed on a GEOS-enabled build on the validated Linux host [10]. This detail matters because the paper claims exact parity with PostGIS on the validated packages, not merely approximate geometric agreement. This does align RTDL's final refine step more closely with the geometric engine family used by PostGIS, but that alignment is intentional: the goal is to remove semantic ambiguity in the final truth predicate, not to erase backend differences in candidate generation.

Parity is checked at the emitted-row level. In the validated artifacts, each system produces a canonical row order, and the final row stream is hashed with SHA-256. This paper therefore includes only cases whose final reported artifacts established exact hash-level agreement across the compared systems.

D. Experimental Scope

This paper uses only final validated artifacts from the repository history. A case is included only if it is already validated and clearly scoped. The paper therefore distinguishes:

- validated package: a stable real-data package or synthetic evaluation surface with completed parity and timing evidence,
- deferred-unavailable: excluded because the public acquisition path is unstable or unavailable,
- not claimed: paper-identical or nationwide statements that the current evidence does not justify.

All reported timings and parity statements in this paper come from the final validated artifacts in the repository history. Earlier provisional or superseded measurements are intentionally excluded.

E. Experimental Environment

Table I summarizes the host environment for the reported four-system results. All four-system runs reported in this paper were executed on the same Linux validation workstation.

F. Packages Used In This Paper

The primary packages used in this paper are:

- County ⋈ Zipcode on top4_tx_ca_ny_pa,
- BlockGroup ⋈ WaterBodies on county2300_s10,
- LKAU ⋈ PKAU on sunshine_tiny,
- LKAU ⋈ PKAU overlay-seed evaluation.

These packages were validated on the same Linux validation workstation. The LKAU ⋈ PKAU package was staged from live OSM Overpass way geometry and then frozen as the validated Australia slice [11].

G. Overlay-Seed Contract

RTDL's current overlay result is not a materialized polygon geometry. The reported overlay row schema is instead an overlay-seed evaluation with four columns:

- left_polygon_id,
- right_polygon_id,
- requires_lsi,
- requires_pip.

This schema records whether a polygon pair requires line-segment intersection refine, point-in-polygon refine, or both. It is the exact surface compared against PostGIS in the validated overlay package.

VI. Results

A. Current Backend Results on the Strongest Surface

The strongest current performance surface in the project is the long exact-source County ⋈ Zipcode positive-hit PIP workload. This is the surface on which RTDL's mature backends now have the clearest performance story against PostGIS. Table II summarizes the final accepted results on that surface.

Across all rows summarized in Table II, exact row-level parity is preserved. The important interpretive point is that the paper now has two distinct kinds of evidence: a bounded package that anchors the broader v0.1 trust story, and a newer long exact-source surface that carries the strongest current backend-performance claim.

B. Bounded Package Support Rows

The bounded package remains the broader v0.1 trust anchor even though it is no longer the strongest performance surface in the paper. That bounded package covers County ⋈ Zipcode on top4_tx_ca_ny_pa, BlockGroup ⋈ WaterBodies on county2300_s10, and the bounded LKAU ⋈ PKAU package, with the overlay row remaining a seed-generation analogue rather than full polygon materialization. These rows matter because they provide the most diverse multi-system package closure, not because they contain the best current backend timings.

C. Lakes/Parks Package

RTDL also reports a lakes/parks package on Australia:

The LKAU package is intentionally small and derived from live OSM way geometry. It is not a continent-scale Australia-family study. However, it is sufficient to support the first validated four-system overlay-seed result reported here. The corresponding native-oracle times are 1.9138 s for LSI, 0.0915 s for PIP, and 5.1447 s for the overlay-seed evaluation.

D. Scalability Studies

The RayJoin Section 5.6 style scalability surfaces are represented as controlled synthetic studies:

- RayJoin Figure 13: LSI query time and throughput under synthetic scaling,
- RayJoin Figure 14: PIP query time and throughput under synthetic scaling.

These are not paper-scale reproductions. They remain valuable because they preserve the experiment structure while staying within a repeatable execution budget.

In the reported package, these studies are driven by deterministic synthetic scaling generators rather than the original full RayJoin data volume. They preserve the same experiment shape—query time and throughput under controlled scale-up—while remaining within the validated execution budget of the current project.

In both studies, the qualitative trend is the one the paper needs to preserve: query time rises with scale, throughput drops as candidate sets thicken, and the probe distribution visibly changes the curve shape. These figures are therefore useful as structural studies even though they are not paper-scale reproductions of the original RayJoin data volume.

E. Performance Interpretation

Figure 3 summarizes the RTDL Figure 15-style overlay speedup study. As with the rest of the overlay surface in this paper, it reflects RTDL's current overlay-seed contract rather than full polygon overlay materialization.

The large Embree-over-oracle speedup on the overlay-seed slice should not be read as a mysterious anomaly. The native oracle is a correctness-oriented exact reference without RT backend acceleration or the same prepared/runtime reuse used by the mature Embree path, so the comparison is most useful as a systems story about what acceleration and runtime specialization recover from a clean semantic baseline.

The performance results in Table II make two points. First, timing boundaries matter. Prepared execution, repeated raw-input calls, and bounded package validation are all honest measurements, but they are not the same measurement and should not be merged into one headline number. Second, RTDL's current backend story is now stronger than the older bounded package table alone would suggest. On the accepted long exact-source County ⋈ Zipcode positive-hit PIP surface, both Embree and OptiX beat PostGIS on prepared execution, and both also beat PostGIS once repeated raw-input execution benefits from runtime-owned reuse. Vulkan is now parity-clean on the same surface, but it remains slower than PostGIS and the other two mature RTDL backends.

The remaining asymmetry is cold-start cost. Embree already has a strong first-call story on the accepted long

TABLE II

Current accepted long exact-source County ⋈ Zipcode positive-hit PIP results. These rows consolidate the final reviewed backend packages rather than a single synchronized rerun, so the PostGIS column is reported as the corresponding accepted range for each boundary.

Boundary	PostGIS (s)	Embree (s)	OptiX (s)	Vulkan (s)	Reading
Prepared exact-source	3.02–3.40	1.04–1.44	2.13–2.54	6.14–6.16	Embree and OptiX faster; Vulkan slower
Repeated raw-input (best repeated)	3.09–3.21	1.09	2.11	6.71	Embree and OptiX faster after runtime reuse
Repeated raw-input (first call)	3.13–3.58	1.96	7.05	16.14	Embree faster; OptiX and Vulkan still pay larger cold-start cost

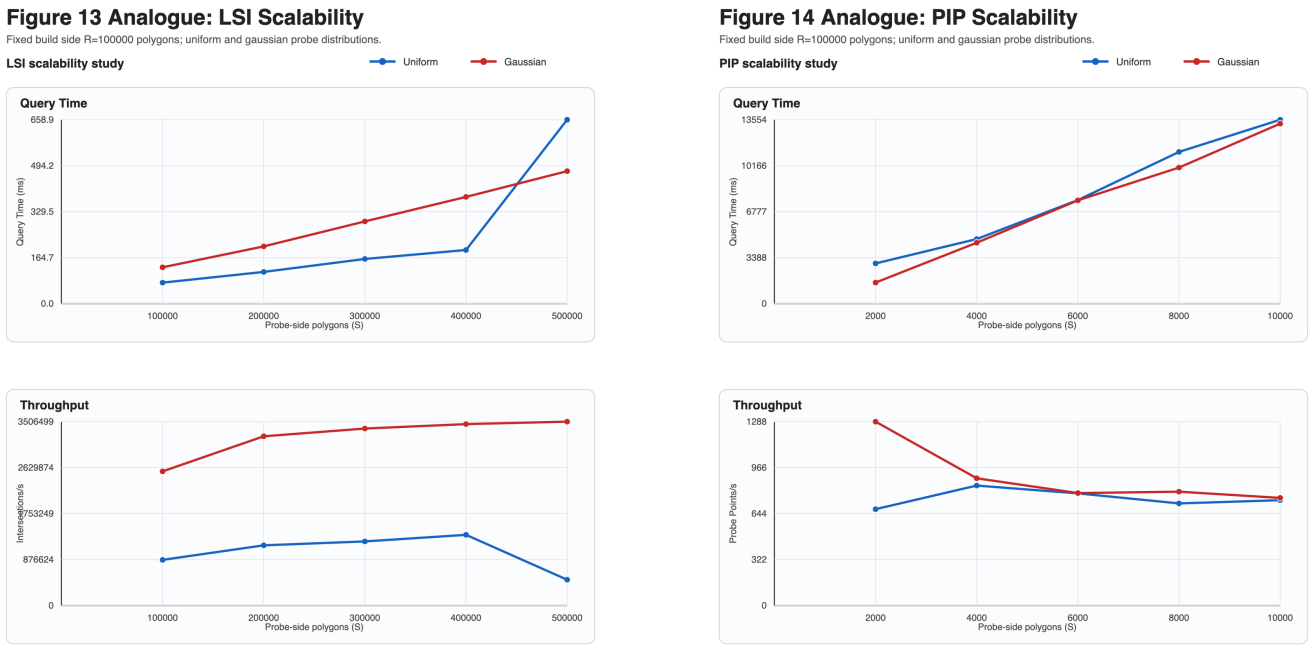

Fig. 2. Controlled studies corresponding to the RayJoin Figure 13 (left, LSI scalability) and Figure 14 (right, PIP scalability) surfaces. Each workload is summarized as a two-panel figure with query time and throughput, and each panel overlays the uniform and gaussian probe distributions used in the bounded analogue study.

TABLE III

Validated LKAU ⋈ PKAU package on sunshine_tiny.

Workload	Rows / Hits	PostGIS (s)	Embree (s)	OptiX (s)
LSI	15	0.0623	2.1286	0.5072
PIP	73920 (22 hits)	0.0041	0.0571	0.3844
Overlay-seed	73920	0.2285	0.0850	0.5674

row, whereas OptiX and Vulkan still pay more one-time setup on the first ordinary raw-input call. That distinction is important because it shows that the main remaining performance question for RTDL is not whether ray-tracing backends can win at all, but how much of the preparation and launch overhead the runtime can hide automatically without making the language harder to use.

VII. Relationship to RayJoin

RayJoin provides the primary workload and evaluation reference for this paper. RTDL does not attempt to replace RayJoin's contribution; instead, it asks a different question: can the same workload family be expressed through a programmable DSL/runtime surface while preserving exact semantics across multiple execution backends? The answer reported here is deliberately scoped. We mirror RayJoin's major workload categories and evaluation surfaces, but we restrict the paper to stable, reproducible packages whose results can be checked both against an internal oracle and, where appropriate, against PostGIS. Unavailable or unstable dataset families are deferred rather than used to support stronger reproduction claims than the evidence allows.

A. Workload Surface

The RTDL evaluation in this paper covers the same workload family categories that matter to RayJoin: LSI,

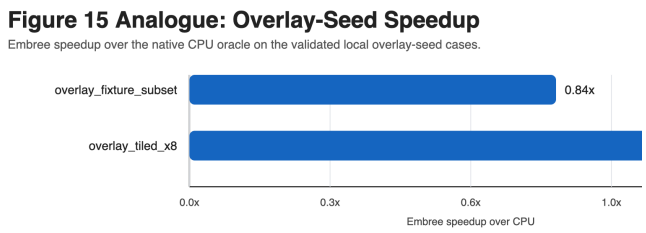

Fig. 3. Overlay-seed speedup study corresponding to the RayJoin Figure 15-style comparison. The plotted bars report Embree speedup over the native CPU oracle for the validated local overlay-seed cases; the study reflects RTDL's current overlay-seed semantics rather than full polygon overlay output geometry.

PIP, and overlay composition.

B. Artifact Structure

The reported evaluation covers:

- Table 3-style package rows,
- Figure 13-style LSI scalability studies (Figure 2, left),
- Figure 14-style PIP scalability studies (Figure 2, right),
- Table 4-style overlay-seed rows,
- Figure 15-style overlay-seed speedup comparisons (Figure 3).

C. Key Differences from the Original Paper

The current RTDL evaluation differs from the original RayJoin paper in four main ways:

1) It is package-scale rather than paper-scale.
2) It relies on controlled synthetic studies where exact paper data is unavailable.
3) It uses PostGIS as an external correctness reference, which RayJoin did not use in the same way.
4) The current overlay surface is a seed-generation study, not full polygon overlay materialization.

VIII. Limitations

This work is intentionally explicit about its limitations.

First, this is not a paper-identical reproduction. Several continent-scale lakes/parks families remain deferred because their public acquisition paths are unstable or unavailable. In the current evaluation these deferred families are:

- LKAF ⋈ PKAF,
- LKAS ⋈ PKAS,
- LKEU ⋈ PKEU,
- LKNA ⋈ PKNA,
- LKSA ⋈ PKSA.

Second, the practical substitute for the paper's Block ⋈ Water row is the county2300_s10 package over BlockGroup and WaterBodies. This scoping qualifier is material to interpretation.

Third, only LKAU ⋈ PKAU currently has four-system overlay-seed results. County/Zipcode and BlockGroup/WaterBodies still provide useful overlay context from earlier Embree work, but not the same four-system overlay evidence.

Fourth, Vulkan is not part of the paper's main performance claim. It is included only as a parity-clean but slower supporting backend on the accepted long exact-source surface.

IX. Conclusion

RTDL now supports an externally checked, multi-backend evaluation of the RayJoin workload family that is substantial enough to analyze in paper form. The contribution of this work is not a claim of full paper-identical reproduction. It is a technically grounded evaluation based on stable real-data results, controlled scaling studies, external PostGIS checks, and exact row-level agreement across the reported systems. On the accepted long exact-source County ⋈ Zipcode positive-hit PIP surface, Embree and OptiX now outperform PostGIS on the reported prepared and repeated-call boundaries, while Vulkan remains parity-clean but slower. Across the bounded package and the long exact-source backend surface, the reported rows preserve exact hash-level emitted-row agreement. That evaluation provides a defensible baseline for subsequent workload expansion, including deferred dataset families, stronger Vulkan maturation, and fuller overlay materialization, without overstating what the current evidence proves.

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